IDENTIFYING INFORMATION:

NAME: Shoele, Kourosh

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-2810-0065

POSITION TITLE: Associate Professor

PRIMARY ORGANIZATION AND LOCATION: Florida State University, Tallahassee, Florida, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of California, San Diego, La Jolla, California, United States	PHD	09/2006 - 08/2011	Structural Engineering
Sharif University of Technology, Tehran, Not Applicable, N/A, Iran	MS	09/2003 - 08/2006	Structural Engineering
University of Shiraz, Shiraz, Not Applicable, N/A, Iran	BS	09/1999 - 08/2003	Civil and Environmental Engineering

Appointments and Positions

2023 - present	Associate Professor , Florida State University, Tallahassee, Florida, United States
2016 - 2023	Assistance Professor , Florida State University, Tallahassee, Florida, United States
2015 - 2016	Research Scientist , Johns Hopkins University, Baltimore, Maryland, United States
2013 - 2015	Postdoctoral Researcher, Johns Hopkins University, Baltimore, Maryland, United States
2011 - 2013	Project Research Engineer, RE-Vision Consulting LLC, Sacramento, California, United States
2011 - 2011	Postdoctoral Fellow, University of California, San Diego, La Jolla, California, United States

Products

1. Ye Z, Estebe C, Liu Y, Vahab M, Huang Z, Sussman M, Moradikazerouni A, Shoele K, Lian Y, Ohta M, Others. An Improved Coupled Level Set and Continuous Moment-of-Fluid Method for Simulating Multiphase Flows with Phase Change. Communications on Applied Mathematics and Computation. 2024; 6(2):1034-1069.
2. Maurya G, Liu Y, Sussman M, Shoele K. Drop transmission after the impact on woven fabrics. International Journal of Multiphase Flow. 2024; 179:104909.
3. Vahab M, Murphy D, Shoele K. Fluid dynamics of frozen precipitation at the air–water interface. Journal of Fluid Mechanics. 2022; 933:A36.
4. Liu Y, Sussman M, Lian Y, Hussaini M, Vahab M, Shoele K. A novel supermesh method for computing solutions to the multi-material Stefan problem with complex deforming interfaces and microstructure. Journal of Scientific Computing. 2022; 91(1):19.

5. Vahab M, Sussman M, Shoele K. Fluid-structure interaction of thin flexible bodies in multi-material multi-phase systems. Journal of Computational Physics. 2021; 429:110008.
6. Wang T, Shoele K. Koopman-based model predictive control with morphing surface: Regulating the flutter response of a foil with an active flap. Physical Review Fluids. 2024; 9(1):014702.
7. Moradikazerouni A, Solano T, Sussman M, Shoele K. Simulation of Natural Convection in Two-Phase Cryogenic Tanks Using Sparse Identification of Nonlinear Dynamics. ; c2023.
8. Shahriar A, Kumar R, Shoele K. Vortex dynamics of axisymmetric cones at high angles of attack. Theoretical and Computational Fluid Dynamics. 2023; 37(3):337-356.
9. Shahriar A, Shoele K. Nonlinear shock-induced flutter of a compliant panel using a fully coupled fluid-thermal-structure interaction model. Journal of Fluids and Structures. 2024; 124:104047.
10. Shahriar A, Shoele K. Fully coupled aero-thermo-elastic analysis of shock-wave and turbulent boundary layer interactions. ; c2024.
11. Tripathi A, Gustavsson J, Shoele K, Kumar R. Effect of shock impingement location on the fluid–structure interactions over a compliant panel. Shock Waves. 2024; :1-19.
12. Shoele K. Hybrid wave/current energy harvesting with a flexible piezoelectric plate. Journal of Fluid Mechanics. 2023; 968:A31.
13. Wang T, Shoele K. Kernel mode decomposition for time-frequency localization of transient flow: The formation of a laminar separation bubble. Physical Review Fluids. 2023; 8(6):064401.
14. Mohammadigoushki H, Shoele K. Cavitation Rheology of Model Yield Stress Fluids Based on Carbopol. Langmuir. 2023.
15. Tripathi A, Sheehan M, Gustavsson J, Shoele K, Kumar R. Effect of panel compliance on shockwave boundary layer interaction at mach 2. ; c2023.
16. Wang T, Shoele K. Identification of transient modes during formation and detachment of a laminar separation bubble using kernel mode decomposition. ; c2021.
17. Ojo O, Wang Y, Erturk A, Shoele K. Aspect ratio-dependent hysteresis response of a heavy inverted flag. Journal of Fluid Mechanics. 2022; 942.

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Shoele, Kourosh in SciENcv on 2026-04-22 23:05:02